

STATE OF MICHIGAN



NATURAL RESOURCES COMMISSION

MARLENE J. FLUHARTY
GORDON E. GUYER
ELLWOOD A. MATTSON
O. STEWART MYERS
RAYMOND POUPORE

JOHN ENGLER, Governor

DEPARTMENT OF NATURAL RESOURCES

STEVENS T. MASON BUILDING
P.O. BOX 30028
LANSING, MI 48909

DAVID F. HALES, Director

May 13, 1991

Mr. Gary Fuller, Director
Wexford County DPW
3161 South Lake Mitchell Drive
Cadillac, Michigan 49601

Dear Mr. Fuller:

Enclosed is Determination of Permit Exemption, GWE - 0337, issued by the Michigan Water Resources Commission at its April 25, 1991 meeting. The Determination provides for the discharge by Wexford County Landfill of up to 108,000 gallons per day of treated pump test waters.

Please review carefully the conditions of the Determination. In particular, please notice that any change in the discharge will require a new Determination by the Commission.

Questions concerning this Determination can be directed to the Groundwater Section, Waste Management Division (telephone 517-373-8148), or the Waste Management Division Cadillac District Office, (telephone 616-775-9728).

Sincerely,

Alan J. Howard acting

Alan J. Howard
Assistant Executive Secretary
Water Resources Commission

Enclosure

cc: District Health Department #1 (Wexford County)
Cedar Creek Township Supervisor
Cadillac District Office

STATE OF MICHIGAN
DEPARTMENT OF NATURAL RESOURCES
WATER RESOURCES COMMISSION

In the matter of: Wexford County Department of Public Works
3161 South Lake Mitchell Drive
Cadillac, Michigan 49601

DETERMINATION OF PERMIT EXEMPTION

Determination No. GWE - 0337

IT IS THE EXPRESS FINDING OF FACT OF THE MICHIGAN WATER RESOURCES COMMISSION THAT:

1. Pursuant to Rule 2206 (R 323.2206) of the Part 22 Rules of the General Rules of the Water Resources Commission: Groundwater Quality, discharges into the groundwaters of the state are regulated by permits issued in accordance with sections 5, 7(1), and 8(b) of the Michigan Water Resources Commission Act (1929 PA 245, as amended).
2. Rule 2209 (R 323.2209) of the Groundwater Quality Rules list activities which are exempt from permit, hydrogeological reporting, and groundwater monitoring requirements.
3. On November 11, 1990, the Wexford County Department of Public Works submitted a request for a Commission determination that a discharge to the groundwater from its facility located in the NW 1/4 of Section 34, T23N, R9W, Cedar Creek Township, Wexford County, Michigan, does not require a discharge permit.
4. The request is for the discharge of up to 108,000 gallons per day of treated pump test waters to the groundwaters.
5. Pursuant to Rule 2209(2), other activities may be exempted from a permit on a case-by-case basis, if the criteria that the Commission deems necessary are met.
6. Gary Fuller, an authorized representative of Wexford County Department of Public Works, has certified, in writing, that the discharge meets the requirements of the rule.
7. Staff have reviewed this request and recommend that the Commission determine that Rule 2209(2) applies and the exemption be granted.

AFTER CONSIDERATION OF THIS MATTER, THE MICHIGAN WATER RESOURCES COMMISSION DETERMINES THAT:

1. The discharge of up to 108,000 gallons per day, of treated pump test water as detailed in the request dated November 11, 1990 for discharge to the groundwaters from the Wexford County Department of Public Works facility located in the NW 1/4 of Section 34, T23N, R9W, meets the requirements of Rule 2209(2).
2. The discharge of treated purged groundwater in excess of 108,000 gallons per day is expressly prohibited.
3. Hydrocarbon contaminants in pump test waters shall be treated to less than 1.0 ppb. Inorganic contaminants shall be treated to levels equal to or less than background.
4. The discharge of any other water or wastewater to the waters of the state without a discharge permit is expressly prohibited.
5. Construction, operation and maintenance of the wastewater treatment and disposal system shall be in accordance with plans and specifications submitted to and approved by the Cadillac District Office of the Waste Management Division of the Michigan Department of Natural Resources prior to discharge.
6. This determination of permit exemption and the authorization to discharge shall expire on September 1, 1991.
7. Provided all the conditions set forth herein are met, no discharge permit under Act 245, P.A. 1929, as amended, is required for this discharge.

The Commission retains jurisdiction to modify this action or to enter any further actions as the facts and circumstances may warrant.

THIS DETERMINATION approved for issuance this 25th day of April 1991, by the Water Resources Commission, to be signed on behalf of the Commission by the Assistant Executive Secretary, effective May 13, 1991.


Alan J. Howard
Assistant Executive Secretary
Water Resources Commission

April 1991

Memorandum to the Michigan Water Resources Commission

RE: Request for Determination of Permit Exemption for
Wexford County Department of Public Works

Briefing Statement and History:

The Wexford County Department of Public Works (DPW) operates the Wexford County Landfill located in the NW 1/4 of Section 34, T23N, R9W, Cedar Creek Township, Wexford County, Michigan. On May 16, 1989, the DPW entered into Consent Order No. 641-05-245-05-1989 with the Michigan Department of Natural Resources, in part, to restore the groundwater from any contamination caused by the existing disposal site to background levels. The development of a remedial action plan by the DPW requires that a pump test be conducted to determine on-site aquifer characteristics. All water pumped during the pump test will be held in a lined 3,000,000 gallon holding basin. From the basin the water can be treated and disposed at a controlled rate not to exceed 108,000 gallons per day.

On November 11, 1990, Gary Fuller, Director of the DPW, submitted an exemption request to discharge treated wastewater from the pump test. The request is based on the water being treated to less than detection levels for organics and to levels equal to or less than background for inorganics. The treated water will be disposed of back to a contaminated area at the site. Both the Consent Order and the draft exemption require approval of plans and specifications for the treatment system prior to the discharge taking place.

Conclusions and Recommendations:

Staff recommends issuance of a permit exemption for disposal of up to 108,000 gallons per day of treated pump test water to the groundwater.

I have analyzed and discussed these recommendations with staff and concur.

Joan H. Peck, acting
Alan J. Howard
Assistant Executive Secretary
Water Resources Commission

STATE OF MICHIGAN
DEPARTMENT OF NATURAL RESOURCES
WATER RESOURCES COMMISSION

In the matter of: Wexford County Department of Public Works
3161 South Lake Mitchell Drive
Cadillac, Michigan 49601

DETERMINATION OF PERMIT EXEMPTION

Determination No. _____

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2. Rule 2209 (R 323.2209) of the Groundwater Quality Rules list activities which are exempt from permit, hydrogeological reporting, and groundwater monitoring requirements.
3. On November 11, 1990, the Wexford County Department of Public Works submitted a request for a Commission determination that a discharge to the groundwater from its facility located in the NW 1/4 of Section 34, T23N, R9W, Cedar Creek Township, Wexford County, Michigan, does not require a discharge permit.
4. The request is for the discharge of up to 108,000 gallons per day of treated pump test waters to the groundwaters.
5. Pursuant to Rule 2209(2), other activities may be exempted from a permit on a case-by-case basis, if the criteria that the Commission deems necessary are met.
6. Gary Fuller, an authorized representative of Wexford County Department of Public Works, has certified, in writing, that the discharge meets the requirements of the rule.
7. Staff have reviewed this request and recommend that the Commission determine that Rule 2209(2) applies and the exemption be granted.

AFTER CONSIDERATION OF THIS MATTER, THE MICHIGAN WATER RESOURCES COMMISSION DETERMINES THAT:

1. The discharge of up to 108,000 gallons per day, of treated pump test water as detailed in the request dated November 11, 1990 for discharge to the groundwaters from the Wexford County Department of Public Works facility located in the NW 1/4 of Section 34, T23N, R9W, meets the requirements of Rule 2209(2).
2. The discharge of treated purged groundwater in excess of 108,000 gallons per day is expressly prohibited.
3. Hydrocarbon contaminants in pump test waters shall be treated to less than 1.0 ppb. Inorganic contaminants shall be treated to levels equal to or less than background.
4. The discharge of any other water or wastewater to the waters of the state without a discharge permit is expressly prohibited.
5. Construction, operation and maintenance of the wastewater treatment and disposal system shall be in accordance with plans and specifications submitted to and approved by the Cadillac District Office of the Waste Management Division of the Michigan Department of Natural Resources prior to discharge.
6. This determination of permit exemption and the authorization to discharge shall expire on September 1, 1991.
7. Provided all the conditions set forth herein are met, no discharge permit under Act 245, P.A. 1929, as amended, is required for this discharge.

The Commission retains jurisdiction to modify this action or to enter any further actions as the facts and circumstances may warrant.

THIS DETERMINATION approved for issuance this _____ day of _____, by the Water Resources Commission, to be signed on behalf of the Commission by the Assistant Executive Secretary, effective _____.

Alan J. Howard
Assistant Executive Secretary
Water Resources Commission



MICHIGAN
DEPARTMENT
OF PUBLIC
HEALTH

MEMORANDUM

DATE: April 30, 1991

TO: Tom Weston
Groundwater Permits Section
Waste Management Division
Department of Natural Resources

FROM: Patrick L. Cook *P. Cook*
Area Engineer
Division of Water Supply
Department of Public Health

SUBJECT: Groundwater Discharge Permit Application for Wexford County
Landfill, Cedar Creek Township, Wexford County

The community, groundwater, and non-community sections of the Division of Water Supply along with the Division of Environmental Health have reviewed the above referenced permit application. No adverse impacts on nearby water supply wells were noted with respect to this proposed discharge.

PLC:lw

cc: James K. Cleland

cc: Thomas Hoogerhyde

cc: District Health Department No. 1 (Wexford)

RECEIVED

MAY 07 1991

Waste Management
Division



MEMORANDUM

To: *McEwan, Ratliff* *Gohike*

Date: 4-22-91

From: *Pat Cook*
District Engineer
Area

Subject: Permit Application - *Wexford County Landfill*

The attached permit application needs to be reviewed to determine if there is any potential impact on adjacent drinking water wells. Please review this application with regard to wells under your jurisdiction. After your review, check the appropriate response below, initial this memo, and pass it along. Let's try to complete all the reviews by 4-30-90. The last person should return this form to me.

Municipal Water Supply Wells

☒ No concerns or comments *Cook PE*

☐ Not enough information provided to comment

☐ Comments to be forwarded directly to DNR

☐ Comments attached*

Type II Public Water Supply Wells

☒ No concerns or comments *McEwan*

☐ Not enough information to comment

☐ Comments to be forwarded directly to DNR

☐ Comments attached*

Private Drinking Water Supply Wells

☒ No concerns or comments *Ratliff*

☐ Not enough information to comment

☐ Comments to be forwarded directly to DNR

☐ Comments attached*

Mobile Home Park and Campground Wells

☒ No concerns or comments *Gohike*

☐ Not enough information to comment

☐ Comments to be forwarded directly to DNR

☐ Comments attached*

*Comments attached to be in finished form that can be forwarded to the Michigan Department of Natural Resources

RECEIVED
MICH. DEPT. OF PUBLIC HEALTH
APR 22 1991

BUREAU OF ENVIRONMENTAL
AND OCCUPATIONAL HEALTH



JOHN ENGLER, Governor

DEPARTMENT OF NATURAL RESOURCES

STEVENS T. MASON BUILDING
P.O. BOX 30028
LANSING, MI 48909

DAVID F. HALES, Director

NATURAL RESOURCES COMMISSION

MARLENE J. FLUHARTY
GORDON E. GUYER
ELLWOOD A. MATTSON
O. STEWART MYERS
RAYMOND POUPORE

April 17, 1991

Mr. Gary Fuller, Director
Wexford County DPW
3161 South Lake Mitchell Drive
Cadillac, Michigan 49601

Dear Mr. Fuller:

SUBJECT: Wexford County Landfill Draft Permit Exemption

Enclosed is a draft of the permit exemption prepared under the provisions of Section 8(b), Act 245, Public Acts of 1929, as amended, for the disposal of water from a test well pump test located at the Wexford County Landfill described in your request dated November 9, 1990.

It is the Water Resources Commission's policy that we send you and other interested parties a copy of the draft permit exemption so that you have an opportunity to review it before it is granted in its final form. Comments must be received as soon as possible. Please call Tom Weston of our office (517-373-8123) if you have any comments or questions on the draft permit exemption.

You are cautioned not to proceed with the construction of treatment facilities until the permit exemption is granted and plans for the facilities have been approved. You will be notified when the permit exemption is granted and will be mailed a copy for your records.

Sincerely,

Joan H. Peck, Chief
Groundwater Section
Waste Management Division
517-335-3383

Enclosure

cc: District Health Department #1 (Wexford Co.)
Cadillac District, Waste Management Division
Cedar Creek Township Supervisor
Water Supply Division, MDPH



INTEROFFICE COMMUNICATION

April 17, 1991

TO: Casey Cline, Cadillac District

FROM: Tom Weston
Groundwater Section, Waste Management Division

SUBJECT Review of Draft Permit Exemption Package for
Wexford County Landfill
Cedar Creek Township, Wexford County

Attached is the draft permit exemption for Wexford County Landfill. We intend to present the draft permit at the April 25, 1991 Water Resources Commission Meeting. If you intend to comment would you please call me as soon as possible to briefly discuss your thoughts. Otherwise we will assume no written comments are forthcoming before we take it to the Commission.

Thank you for your assistance.



Attachment

INTEROFFICE COMMUNICATION

April 17, 1991

TO: James Cleland, Chief
Water Supply Division
Michigan Department of Public Health

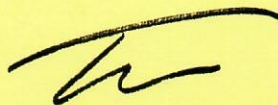
FROM: Tom Weston
Groundwater Permits Section
Waste Management Division
Department of Natural Resources

SUBJECT: Potential Impact on Water Supply Wells
Wexford County Landfill
Cedar Creek Township, Wexford County

Would you please have your staff review the attached briefing memo/fact sheet to determine if there is any potential impact on adjacent drinking water wells and advise us if there are any problems. The proposed permit is being placed on public notice at this time and we intend to take it to the April 25, 1991 Water Resources Commission meeting. Any comments which might affect staff's recommendations or the Commission's decision to issue or deny the permit would be appreciated before that date.

If your staff has any questions on this proposed project the should direct their questions to Catherine (Casey) Cline at our Cadillac District Office (616-775-9728).

Thank you for your consideration.



Attachment: Copy of Draft Permit Exemption

cc: Casey Cline, Cadillac District Office, WMD

NATURAL RESOURCES COMMISSION

MARLENE J. FLUHARTY
GORDON E. GUYER
ELLWOOD A. MATTSON
O. STEWART MYERS
RAYMOND POUPORE

STATE OF MICHIGAN



JOHN ENGLER, Governor

DEPARTMENT OF NATURAL RESOURCES

STEVENS T. MASON BUILDING
P.O. BOX 30028
LANSING, MI 48909

DAVID F. HALES, Director

April 17, 1991

Mr. Gary Fuller
Wexford County DPW
3161 S. Lake Mitchell Drive
Cadillac, Michigan 49601

Dear Applicant:

Subject: April 25, 1991 WRC Meeting Agenda - Wexford County Landfill

Attached is an agenda for the upcoming Michigan Water Resources Commission meeting. The meeting date and location are indicated at the top of the agenda.

You will note that staff has scheduled an item pertaining to your facility for Commission action at this meeting. It is not necessary for you to attend the meeting unless you desire to comment on the proposed action.

Sincerely,

A handwritten signature in cursive script, reading "Scott C. Ross".

Scott C. Ross, Chief
Groundwater Permits Unit
Waste Management Division
517-373-8148

Attachment

NATURAL RESOURCES COMMISSION

MARLENE J. FLUHARTY
GORDON E. GUYER
ELLWOOD A. MATTSON
O. STEWART MYERS
RAYMOND POUPORE

STATE OF MICHIGAN



JOHN ENGLER, Governor

DEPARTMENT OF NATURAL RESOURCES

STEVENS T. MASON BUILDING
P.O. BOX 30028
LANSING, MI 48909

DAVID F. HALES, Director

April 8, 1991

Mr. Gary Fuller
Wexford County Dept. of Public Works
990 One Hundred and Thirty One Road
Manton, Michigan 49663-0100

Dear Mr. Fuller:

Subject: Application for Groundwater Discharge Permit

The exemption for your groundwater pump test discharge is currently scheduled for the April Water Resources Commission meeting. Enclosed is an application for a groundwater discharge permit for the purge water discharge for your groundwater remediation program. If you have any questions about it, please contact me or Thomas Weston at 517-335-4723.

Sincerely,

Kathleen H. Sibbo

Kathleen H. Sibbo, Geologist
Waste Management Division
Groundwater Section
517-373-7144

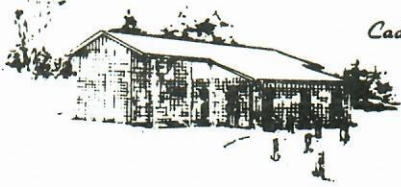
cc: C. Cline, MDNR Cadillac
T. Weston, MDNR Groundwater

Wexford County Department of Public Works

3161 South Lake Mitchell Drive

Cadillac, Michigan 49601

Telephone Area 616-775-0155



April 3, 1991

Ms Kathleen H. Sibb, Geologist
Waste Management Division
Michigan Department of Natural Resources
P.O. Box 30028
Lansing, Michigan 48909

REF: Groundwater Discharge Permit Exemption Application
Wexford County Landfill

Dear Ms Sibb:

We have been advised by Ms Catherine Cline at your Cadillac District office that there is likely to be an additional three (3) months delay in issuance of the above titled permit.

Our application has been in your office several months for processing and any further delay will jeopardize start-up of our groundwater remediation program this year. Additionally, and as critical, are the compliance schedule dates that have been agreed upon in our Consent Order. We are concerned that unreasonable delays by the Michigan Department of Natural Resources in reviewing our proposals and issuing approvals may place us in default of the Order. Therefore, we are requesting that your office explain in writing why we have not been issued an exemption on the subject matter in a timely manner.

It is also our understanding that district office staff has volunteered their time to assist your office in the tasks necessary to expedite permit approval. We strongly encourage you to accept their offer and also extend our willingness to assist you in any way possible.

If you have any questions please do not hesitate to contact this office immediately.

Sincerely,

A large, stylized handwritten signature in dark ink, appearing to read "Gary R. Fuller".

Gary R. Fuller,
Director

RECEIVED

APR 05 1991

GRF/pal

cc: Philip Roycraft
Catherine Cline
William Pearson

Waste Management
Division

RECEIVED

Pearson Drilling & Engineering, Inc.

3985 S. LaChance Rd. — Lake City, MI 49651
(616) 779-0444

FEB 27 1991

Waste Management

February 25, 1991
Division

Ms. Kathleen H. Sibbo
Waste Management Division-MDNR
P.O. Box 30028
Lansing, MI 48909

Re: Wexford County Landfill Groundwater Permit Exemption
Application

Dear Ms. Sibbo,

In response to your January 31, 1991 letter to Mr. Gary Fuller, Director of Wexford County Department of Public Works, I am sending you the enclosed documents which should provide the information you requested.

The enclosed "Hydrogeological Study for the Wexford County Landfill" and "Part II" contain several maps which show the landfill location and ownership of surrounding land in the immediate vicinity of the landfill. Also a site sketch showing the holding lagoon and the proposed discharge area has been enclosed. The studies also contain well logs for nineteen monitor wells which have been installed at this site. Well locations are shown on the maps which can be found in the map pockets of each study.

I have also included a copy of the "Aquifer Test Work Plan" which details our anticipated plan of action for the pump test. The pump test discharge water will be stored in a 20 mil PVC lined lagoon. After the pump test has been completed, the water in the lagoon will be tested and a treatment system will be designed using the test results. It is anticipated that activated carbon filtration or air stripping will be used to treat the water. Storing the pump test water in a temporary lagoon will allow the use of a much smaller treatment system than one with the capacity to treat the proposed test volume of 500 gpm. The anticipated discharge volume of the temporary treatment system for treating the lagoon water, is less than 75 gallons per minute.

Discharge to the ground surface at a rate of 75 gpm or less should have little effect on the aquifer which is greater than 150 feet below the land surface. Any source of contamination contributing to the plume is 300 feet or greater downgradient of the discharge area. The permit exemption request is for the discharge of pump test water only.

If you have any questions or seek additional information, please feel free to contact myself at 616-779-0444.

Sincerely,


William F. Pearson, P.E.

xc. G. Fuller, Wexford DPW
C. Cline, MDNR Cadillac District Office

January 31, 1991

Mr. Gary Fuller
Wexford County Dept. of Public Works
990 One Hundred and Thirty One Road
Manton, Michigan 49663-0100

Dear Mr. Fuller:

SUBJECT: Groundwater Permit Exemption Application
Wexford County Landfill

I have reviewed the geological aspects of your application and have the following list of questions and deficiencies:

1. There is no location information provided, other than the site is somewhere in Wexford County by US 131. Please provide a map showing the site location and its boundaries. This can be done on a topographic map.
2. You state in your application that no wells or residences are within 1/2 mile of the site. However, no information is provided to verify this. Please provide a map showing the locations of the nearest wells (this can be done on the same map as your site location).
3. You state that the treated water will be discharged to a location at the south end of the landfill, but don't say how it will be discharged, or at what rate. Please provide a description of the proposed treatment and disposal systems, including the discharge rate.
4. In your application, you state that the discharge area soils are sand but provide no site information to support this or to demonstrate that the site is permeable enough for your discharge method and rate to work. Please provide site soils classifications and characteristics, logs of on-site borings, and permeability test results for the proposed discharge area.
5. No information about the depth to groundwater and location of aquifers and protective layers was provided. Please provide a representative selection of area well logs with each log correlated with its location on a map.

Mr. Gary Fuller
Page 2
January 31, 1991

6. No detailed information on groundwater flow and plume venting was provided. Please provide this with a map (can also be included on the map discussed above).
7. In your application, you claim that even though the discharge area is upgradient of the main plume, the discharge will not adversely affect the plume. However, no mounding calculations were submitted to substantiate this.
8. It is unclear from your description, whether you are applying only for one pump test discharge or also for the discharge of treated purge water. Please be aware that if there is to be groundwater discharge of treated purge water, a permit will be needed.

Please provide the above information by March 11, 1991. Your application cannot be processed without it. If you have any questions, feel free to call me.

Sincerely,

Kathleen H. Sibb, Geologist
Waste Management Division
Groundwater Section
(517) 373-7144

cc: W. Pearson, Pearson Drilling & Eng., Inc.
T. Weston, MDNR Groundwater
C. Cline, MDNR Cadillac

January 15, 1991

TO: Tom Weston
Permits Unit, WMD

FROM: Kathleen Sibb
Hydrogeologic Review Unit, WMD *KAS*

SUBJECT: Wexford County Landfill

I have reviewed this exemption application and have found it to be extremely deficient. Little site or area information is provided, and even this is unsubstantiated. My list of questions and deficiencies is as follows:

1. There is no location information provided, other than the site is somewhere in Wexford County. There is one general site map, but it shows nothing except US 131 outside the landfill boundaries.
 2. A proposed discharge area is located in the south part of the property, but this area protrudes beyond the rectangular landfill boundaries. Is this now landfill property? What lies around it?
 3. Pump test water will be discharged to a lagoon lined with a 20 mil PVC liner. Is 20 mil thick enough to store water with solvents for a time?
 4. The applicant states that the lagoon water will be treated but doesn't say how. Supposedly, the lagoon storage will allow for sampling and for design of the treatment system. Shouldn't this be done in advance, particularly since they have already been sampling the plume and should know what contaminants and concentrations to expect?
 5. The applicant also states that the treated water will be discharged to a location at the south end of the landfill, but doesn't say how it will be discharged, or at what rate.
 6. The applicant states that no wells or residences are within 1/2 mile of the site. No map information is provided to verify this.
 7. The applicant states that the discharge area soils are sand on a fairly level surface to promote seepage into the ground and prevent erosion. No information to support this was provided. No soils information, boring or well logs, or permeability test results were provided.
- collected*

Tom Weston
Page 2
January 15, 1991

8. The applicant claims that even though the discharge area is upgradient both of a filled area that has shown low organics contamination and the main plume, the discharge will not adversely affect the plume. This is because mounding should occur upgradient of any contaminant source. However, no mounding calculations or plume location map was submitted to substantiate this.
9. No information about the depth to groundwater and location of aquifers and protective layers was provided. We don't know if they are discharging to a usable aquifer or not.
10. No water quality data was provided.
11. No detailed information on groundwater flow and plume venting was provided. From the comments in the text, apparently the groundwater is flowing from south to north.

Because this application is so deficient, but also needs to be processed fairly quickly, perhaps we should call Casey Cline to discuss it once you've had a chance to review it.

MICHIGAN DEPARTMENT OF NATURAL RESOURCES

INTEROFFICE COMMUNICATION

CADILLAC DISTRICT OFFICE
8015 Mackinaw Trail
Cadillac, Michigan 49601

December 6, 1990

TO: Jim Janiczek, Chief, Hydrogeologic Unit
Waste Management Division

FROM: Casey Cline, Cadillac District Office
Waste Management Division

SUBJECT: Submittal of Groundwater Discharge Permit Exemption
from Wexford County Landfill

Jim, please note the attached permit exemption request. I have additional file information which you may wish to review. For the sake of saving on copying costs, perhaps you (or whomever handles this exemption) can call me and discuss this site and request further items in the file. You are also free to call Gary Fuller, of Wexford County DPW, who prefers being contacted before you contact their consultant directly.

This site is operating under a consent agreement and a current Act 641 operating license. They have documented groundwater contamination and have determined the general extent of the groundwater contamination plume. Part of this phase of investigation will include a pump test and treatability test and further sampling of the outlying wells to confirm the limits of the inorganic and organic plume at this site. I understand that this exemption may be given a high priority given that it is under a consent agreement as a clean-up and is an exemption and not a permit application.

Casey Cline

CC/kjm
cc: Phil Roycraft
Jim Sygo
Gary Fuller
Bill Pearson

RECEIVED

DEC 18 1990

Waste Management
Division

REQUEST FOR DETERMINATION OF PERMIT EXEMPTION
FOR A GROUNDWATER DISCHARGE

DEC 18 1991

WGMT-PT

General Rules of the Water Resources Commission - Groundwater Quality provide:

Rule 2209.(1) The following activities do not require a permit from the commission or a hydrogeological report or groundwater monitoring, except as may be required by the commission on a case-by-case basis, where such activities are or may become injurious to the protected uses of a usable aquifer:

(a) Disposal of sanitary wastes in volumes of less than 10,000 gallons per day through the use of septic tank and ground disposal systems approved by the local county and district health departments that are certified by the commission.

(b) Controlled application of dust-suppressant chemicals used with normally accepted or regulated practices.

(c) Controlled application of de-icing chemicals used with normally accepted or regulated practices.

(d) Controlled chemical applications for natural resource and right-of-way programs used with normally accepted or regulated practices.

(e) Controlled application of chemicals for domestic purposes used with normally accepted or regulated practices.

(f) Controlled application of chemicals for agricultural and silvicultural use by normally accepted or regulated practices.

(g) Disposal of non-contact cooling water, not treated by the addition of chemicals, in volumes established at the discretion of the commission.

(h) Retention of storm water runoff in surface impoundments or surface waterways.

(2) Other activities may be excluded as determined by the commission on a case-by-case basis, if the criteria, limitations, or conditions that the commission deems necessary are met.

I, Gary Fuller, Director
(Name of Authorized Representative) (Title)

of Wexford County Department of Public Works
(Company, Unit of Government, etc.)

3161 S. Lake Mitchell Dr. Cadillac, MI 49601
(Address)

request a Determination of Permit Exemption from the Water Resources Commission under Commission Rule 2209.

I certify the discharge for which I am requesting a permit exemption meets the requirements of Rule 2209 2
(1) or (2) letter (a) thru (h) as appropriate

Please describe discharge (including location) in space below: ,

The proposed discharge will be treated water generated from a pump test. The pump test is necessary to complete groundwater remediation at the Wexford County Landfill.

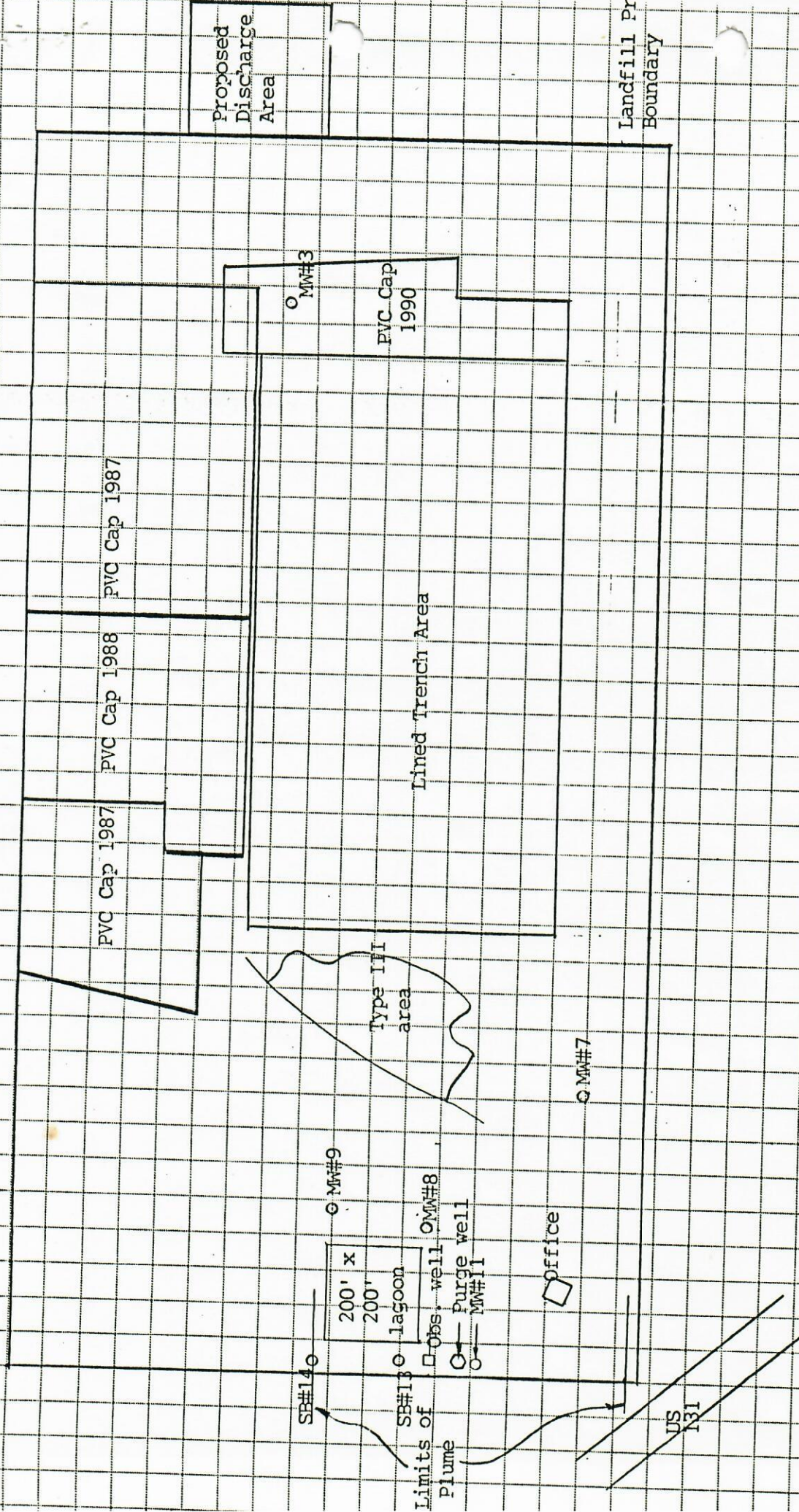
The pump test discharge will be temporally stored in a lined lagoon. From the lagoon, the water will be treated and discharged to the ground surface. The discharge location is 250 feet upgradient of any sources of contamination. Soils at the discharge area are sand. The ground surface at the discharge site is relatively level. The level ground and sandy soil will allow the water to seep into the ground. Run-off or erosion will not be a problem.

Total volume of the anticipated discharge will not exceed 3,000,000 gallons.

The discharge location is well isolated. No water supply wells or residences exist within 1/2 mile of the site.

COMPLETE AND RETURN TO:
Groundwater Permits Section
Waste Management Division
P. O. Box 30028
Lansing, MI 48909


Signature
11/9/90
Date



Lagoon Volume = 3,000,000 gallons
 400,000 cu. ft.

Natly Silo

Jim J. Kneezel

SUBJECT:

Deaford County, Tenn. Hill

OUR JOB NO.

DATE OF MEMO

01-10-90

MESSAGE

Please review the appropriateness of the attached
examination request, which is for a pump test necessary to
ensure a remedial action. Clark & Kneezel, Inc. but I
believe this needs expedited review. If approvable, please
note this in a memo which includes a very brief summary
of the geology and hydrogeology of the site.

SENDER — DO NOT WRITE BELOW THIS LINE

SIGNED

REPLY

SIGNED

DATE

ORIGINAL

SENDER — Retain part 2 for your follow-up, send parts 1 and 3 to addressee
RECIPIENT — Retain part 1 and return part 3

Status Sheet

Consent Order #641-05-245-05-1989

Facility: Wexford County Landfill

	<u>CONDITIONS</u>	<u>Due Date</u>	<u>Completed</u>
1.	<u>Closure</u>		
8.	Discontinue landfilling unlined areas	11/01/89	01/11/90
11.	Certify closure of unlined areas	06/01/90	
2.	<u>Hydrogeological Study and Groundwater Monitoring</u>		
14.	Implement study plan	06/01/89	06/01/89
15.	Complete study	11/01/89	11/01/89*
18.	Submit remedial action plan	02/01/90	
3.	<u>Licensure of Double-Lined Areas</u>		
27.	Initiate construction	06/01/89	06/01/89
28.	Complete construction Submit operating license application	08/01/89	08/01/89
31.	Apply for title to land	07/25/89	07/24/89
32.	Submit restrictive deed	10/25/89	12/15/89

*Additional phases of study will be necessary.

STATE OF MICHIGAN



JAMES J. BLANCHARD, Governor

DEPARTMENT OF NATURAL RESOURCES

STEVENS T. MASON BUILDING
P.O. BOX 30028
LANSING, MI 48909

DAVID F. HALES, Director

NATURAL RESOURCES COMMISSION

THOMAS J. ANDERSON
MARLENE J. FLUHARTY
GORDON E. GUYER
KERRY KAMMER
O. STEWART MYERS
DAVID D. OLSON
RAYMOND POUPORE

May 30, 1989

Mr. Gary Fuller, Director
Wexford County Department of Public Works
3161 S. Lake Mitchell Drive
Cadillac, MI 49601

Dear Mr. Fuller:

Enclosed please find a fully executed original of the Consent Order between the Wexford County Department of Public Works and the Department of Natural Resources. The Consent Order became effective on May 25, 1989, the date of the Director's signature.

Also enclosed is a handwritten note from the Director to Mr. Frank Ziska expressing his appreciation for the County's willingness to resolve the problems at the landfill through entry of this Consent Order. I share in the Director's appreciation of the County's efforts and look forward to completion of the compliance program outlined in the Consent Order.

Sincerely,

A handwritten signature in cursive script that reads "Dennis M. Drake".

Dennis M. Drake, Chief
Compliance and Enforcement Section
Waste Management Division
517-373-7938

Enclosure

cc: Mr. Phil Roycraft, DNR, w/enclosure

NATURAL RESOURCES COMMISSION

THOMAS J. ANDERSON
MARLENE J. FLUHARTY
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O. STEWART MYERS
DAVID D. OLSON
RAYMOND POUPORE

STATE OF MICHIGAN



JAMES J. BLANCHARD, Governor

DEPARTMENT OF NATURAL RESOURCES

STEVENS T. MASON BUILDING
BOX 30028
LANSING, MI 48909

DAVID F. HALES, Director

*To Frank Ziska
Wexford Co. Board of Public Works*

*I'm pleased to sign the Consent
Order, and I'm very appreciative
of Wexford Co's constructive
and responsible action.*

David Hales



STATE OF MICHIGAN
DEPARTMENT OF NATURAL RESOURCES

In the matter of the
Wexford County Department of Public Works
3161 South Lake Mitchell Drive
Cadillac, Michigan

/ WMD Order No. 641-05-245-05-1989

CONSENT ORDER

The Department of Natural Resources of the State of Michigan (hereinafter "the Department") and the Wexford County Department of Public Works (hereinafter "the County DPW") agree and stipulate to the following:

1. The Director of the Department is the State official charged with the administration and enforcement of the Water Resources Commission Act, 1929 PA 245, as amended (hereinafter the "Water Resources Commission Act"), an act to protect the public health, environment, and the waters of the State, the Solid Waste Management Act, 1978 PA 641, as amended, (hereinafter the "Solid Waste Management Act") an act to provide for the regulation and management of solid waste; and the Thomas J. Anderson, Gordon Rockwell Environmental Protection Act, 1970 PA 127.
2. The County DPW operates a landfill (hereinafter "the disposal site") in T23N, R9W, Section 34, Cedar Creek Township, Wexford County, Michigan, consisting of approximately eighty (80) acres (see Attachment A). Solid waste has been disposed of in approximately fifteen (15) unlined acres of the disposal site (hereinafter "unlined areas"). The unlined areas consist of Phase I, Phase II, and active areas. Phase I has been closed, and capped, but topsoil and seeding have not been completed. Phase II has been closed, capped, and topsoiled but seeding has not been completed. The active areas shown on Attachment A remain open.
3. The County DPW desires to operate Phase III of the disposal site (consisting of approximately 21 acres and adjacent to Phases I and II) as a lined, licensed disposal area. The County DPW desires to operate the southwest corner of the disposal site (consisting of approximately two acres) for inert material.
4. The Department alleges, but the County DPW does not admit, that past landfill operations and other disposal activities at the disposal site has resulted in, and continues to cause, unpermitted discharges to, and resultant contamination of, the groundwaters of the State in violation of the Water Resources Commission Act.
5. For purposes of this Consent Order, the definition of contamination shall be based on levels of constituents which exceed background levels. Background levels of constituents of the groundwater shall be established by sampling and analysis procedures complying with 40 CFR 264.97(d) (1986 Ed.) using nearby monitoring wells that are not affected by any existing plume of contamination. The method for determining background levels of

a constituent of the groundwater in those wells shall be to collect four samples to be analyzed as four separate samples and to obtain a mean value for the constituent in the manner described in 40 CFR 264.97(h) (1986 Ed.). Alternative methods outlined in the Resource Conservation and Recovery Act Groundwater Monitoring Technical Enforcement Guidance Document of September 1986, or other alternative methods proposed by the County DPW, may be used to determine background levels if approved by the Department.

6. In order to resolve any groundwater contamination problems at this site, the County DPW agrees to establish procedures which will minimize the infiltration of rain and surface water into the unlined disposal areas, contain, control, and prevent further contamination of resources, and to fully close the active, unlined disposal areas.
7. The County DPW agrees to comply with all provisions of the Solid Waste Management Act, the Water Resources Commission Act, and all administrative rules promulgated under these acts.

Closure of Unlined Areas

8. The County DPW shall discontinue the landfilling of solid waste within the unlined areas by November 1, 1989.
9. The County DPW has submitted to the Department a complete closure plan for the unlined areas, as part of the engineering plans dated July 13, 1988. This plan was prepared by a registered professional engineer and delineates the extent of all refuse filled areas and the extent to which each has received final cover. The plan also describes, defines, and identifies grading contours, final slopes, surface water drainage patterns, specifications for final cover, gas venting systems, final cover seeding and stabilization, and utilizes a PVC cap for the protective cover.
10. Upon execution of this Consent Order, the Department hereby approves the closure plan.
11. By June 1, 1990, the County DPW shall complete the closure of all unlined areas in accordance with the approved closure plan and shall submit a certification to the Department that the closure of the unlined areas has been completed.

Post-Closure Inspections

12. All completed, capped areas shall be inspected by the County DPW periodically (after heavy rainfall and at least monthly during the first year and quarterly thereafter) to monitor the integrity of the final cover. Reports of the inspections shall be submitted to the Department within 30 days of the end of each calendar quarter and will include the conditions of the site at the time the inspection was made, any actions taken to address problems, and additional actions needed to address the problems. Upon discovery, any areas of erosion from runoff, off-road vehicles, etc., shall be repaired by the County DPW by restoring the thickness of the protective cover followed by application of topsoil and

seeding, as necessary. Post-closure inspections shall extend for a minimum of thirty (30) years after certification of the cap and shall continue for as long as the groundwater monitoring is conducted pursuant to Paragraph 16 of this order.

Disposal Site Hydrogeological Study and Groundwater Monitoring Program

13. The County DPW has submitted to the Department a hydrogeological study plan that is designed to define the horizontal and vertical extent of any plume of contaminated groundwater emanating from the disposal site and to define aquifer characteristics (See Attachment B). The hydrogeological study shall include the installation of a sufficient number of monitoring wells that are deemed necessary by the Department to define, within 100 feet of any existing potable water supply well and within 900 feet of all other areas, the lateral extent of any plume of groundwater contamination. The County DPW shall be responsible for obtaining permission to enter off-site property to place monitoring wells, conduct surveys, take samples, or complete other tasks as may be appropriate to the hydrogeological study. Upon the request of the County DPW, the Department agrees to contact the owners or occupants of off-site properties and explain the need for the County DPW to enter their property in order to comply with the terms and conditions of the consent order. In the event reasonable access is refused, the County DPW shall purchase the property or initiate litigation to acquire access by October 1, 1989. In the event purchase or litigation is necessary to acquire access, the hydrogeological report shall be completed within 120 days of acquisition.
14. By June 1, 1989, the County DPW shall implement the hydrogeological study plan.
15. By November 1, 1989, the County DPW shall complete the hydrogeological study and shall provide the completed results to the Department in the form of a report, together with a work plan for a proposed groundwater monitoring program, as described in Paragraph 16 of this order.
16. The County DPW shall operate a groundwater monitoring program to monitor any plume of contaminated groundwater emanating from the site. In order to monitor any plume of contaminated groundwater, the County DPW shall, by November 1, 1989, submit to the Department a work plan, acceptable to the Department, outlining a proposed groundwater monitoring program to monitor any plume of contamination emanating from the disposal area. This work plan shall identify the number and location of wells, parameters to be analyzed, methods of testing to be used, the duration of sampling, the laboratory(ies) which will analyze the samples, sampling equipment, sample handling procedures, decontamination procedures, chain of custody procedures, and quality assurance/quality control programs outlined in the Resource Conservation and Recovery Act Groundwater Monitoring Technical Enforcement Guidance Document of September 1986. Monitoring shall continue for a sufficient period of time to determine that groundwater has been completely remediated, but in no event shall be less than a period of five (5) years after October 1, 1989. The County DPW shall implement the collection of groundwater samples from all designated monitoring wells (identified in the groundwater monitoring

work plan) within 90 days of the Department's written approval of the work plan.

17. The County DPW shall, at a minimum, collect and analyze all groundwater samples for those inorganics, metals, and organics listed in Attachment C on a quarterly basis for a minimum of two years. If after the first two years, a particular metal or all parameters within an organic scan are not found in the groundwater for four consecutive sampling events, the sampling program may be revised with Department approval to eliminate the requirement for analysis for that particular metal or organic scan on a quarterly basis, provided however that upon the reappearance of any such parameter(s), the Department may require the County DPW to resume sampling for such parameter(s). After two years of groundwater sampling and analysis pursuant to the Department approved groundwater monitoring program, the program may be revised to provide for sampling on a twice-yearly basis. Any revision to the sampling program shall be reviewed and approved by the Department prior to implementation. All sampling results shall be submitted to the Department within thirty (30) days of the end of the calendar quarter.

Groundwater Remedial Action Plan

18. The County DPW shall restore the groundwater from any contamination caused by the existing disposal site to background levels.
19. The County DPW shall submit to the Department by no later than February 1, 1990, for its review and approval, a remedial action plan to restore the groundwater to background levels and to prevent any future contamination of the groundwater from the disposal site. The remedial action plan shall contain all of the following:
 - A. A description of the selected alternative, and a description of the equipment, installations, and permits necessary to implement the plan.
 - B. The date upon which complete applications will be made for any permit necessary to implement the plan.
 - C. The date when necessary equipment will be installed.
 - D. The date upon which remediation will begin.
20. If the Department formally promulgates and establishes, pursuant to the Administrative Procedures Act, MCL 24.201, et seq., an applicable, less restrictive than background standard, that new, less restrictive standard shall then apply to the County DPW's obligation and to the remedial action plan.
21. Within forty-five (45) days of written approval of the remedial action plan by the Department, the County DPW shall implement the same in accordance with the approved schedule for implementation.

Phase III Groundwater Monitoring Program

22. Upon licensure under the Solid Waste Management Act, the County DPW shall operate a groundwater monitoring program to monitor the performance of the Phase III licensed area. In order to monitor the Phase III area, the County DPW shall sample groundwater monitoring wells MW3, MW4, MW6, MW7, and MW8, quarterly and analyze the groundwater for the constituents listed in Attachment C of this order.
23. With the first operating license application for Phase III, the County DPW shall submit to the Department the procedures to be used for monitoring the Phase III area, including parameters to be analyzed, methods of testing to be used, the duration of sampling, the laboratory(ies) which will analyze the samples, sampling equipment, sample handling procedures, decontamination procedures, chain of custody procedures, and quality assurance/quality control programs as outlined in the Resource Conservation and Recovery Act Groundwater Monitoring Technical Enforcement Guidance Document of September 1986. At a minimum, the County DPW procedures shall include the procedures specified below:
- A. The County DPW shall notify the Department at least one (1) month prior to the date it intends to sample any groundwater monitoring well.
 - B. The static water elevation shall be determined by methods giving precision to 1/8 inch or 0.01 foot prior to purging water from the wells for sampling. Measurements shall be made from the top of the casing with the elevation of all casings in the monitoring well system related to a permanent reference point, using United States Geological Survey datum.
 - C. The samples obtained shall be representative of groundwater. To insure a representative sample, the County DPW shall purge a volume of water equal to or greater than three (3) times the amount of water in the well casing or until dry before obtaining a sample for analysis. Wells shall be sampled within 48 hours after purging where recovery rates allow. Where wells are pumped dry during purging, recovery rates shall be determined and samples taken as soon as sufficient recovery occurs.
 - D. Non-dedicated pumps or sampling equipment shall be thoroughly cleaned and rinsed before use in each monitoring well. Procedures shall minimize the potential for cross-contamination and insure that interference with analytical methods does not occur.
 - E. Water pumped from a monitoring well which is contaminated shall only be discharged to a facility licensed under Michigan's Water Resources Commission Act, 1929 PA 245, or other applicable state law.
 - F. All wells shall be clearly labeled, securely capped, and locked when not in use.
 - G. During the monitoring period, the County DPW shall preserve, maintain, rehabilitate and, if necessary, replace all groundwater monitor wells at the disposal site so that they are available and usable by the County DPW and the Department for monitoring purposes.

Prior to undertaking well replacement or repair, the licensee shall obtain Department approval.

24. The County DPW shall collect and analyze all groundwater and leachate samples for those inorganics, metals, and organics shown in Attachment C for a minimum of two years. If after the first two years, a particular metal or all parameters within an organic scan are not found in the groundwater for four consecutive sampling events, the sampling program may be revised with Department approval to eliminate the requirement for analysis for that particular metal or organic scan on a quarterly basis, provided however that upon the reappearance of any such parameter(s), the Department may require the County DPW to resume sampling for such parameter(s). Any revision to the sampling program shall be reviewed and approved by the Department prior to implementation. All sampling results shall be submitted to the Department within thirty (30) days of the calendar quarter.

Licensure of Phase III

25. The County DPW shall not place solid waste or otherwise fill in the Phase III area at the disposal site until an operating license has been granted by the Director of the Department of Natural Resources.
26. The County DPW has submitted engineering plans and specifications for the Phase III area dated July 13, 1988. These engineering plans describe the installation of double liner system in the base of the Phase III area consisting of a primary (top) liner, a leak detection system, and a secondary liner. Upon execution of this Consent Order, the engineering plans and specifications are approved.
27. The County DPW shall by June 1, 1989, initiate construction of the Phase III area.
28. By August 1, 1989, the County DPW shall complete construction of the Phase III area in accordance with approved plans and submit a complete operating license application for all or portions of the Phase III area. A complete operating license application shall include all of the following:
 - A. A written plan, acceptable to the Department, for determining whether a leak has occurred in the primary (top) liner. The plan shall specify that the County DPW will, at a minimum, do all of the following:
 - i. Inspect the leak detection system not less than weekly.
 - ii. Collect any liquid found in the system and measure its volume.
 - iii. Analyze the liquid for those constituents listed in Attachment C.
 - iv. Analyze leachate from the leachate collection system of Phase III not less than quarterly for those constituents listed in Attachment C.

- v. Determine whether a leak has occurred in the primary (top) liner, based on the volume and quality of liquid found in the leak detection system, if any.
 - B. A written contingency plan, acceptable to the Department, outlining the actions the County DPW will take in the event the primary (top) liner fails.
 - C. Other information required of an operating license application by the Solid Waste Management Act.
29. The Director of the Department shall grant an operating license for the Phase III area following submission of a complete operating license application as required by Rule 313(2) of the administrative rules of the Solid Waste Management Act, provided the Department determines that the Phase III area meets all applicable criteria for licensure. The application for an operating license shall include a verification by a registered professional engineer that the Phase III area has been constructed in accordance with the engineering plans approved pursuant to paragraph 26.

Title to Land

30. Title to the parcel of land within the disposal site is currently vested in the State of Michigan.
31. Within 60 days of the execution of this order, the County DPW shall submit, on a form provided by the Department, an application to the Department for the conveyance of the title of the land within the Phase III area and associated active work areas (See Attachment A) to the County DPW.

Restrictive Deed Covenant

32. No conveyance of title, easement, or other interest in the property referred to herein as the Phase III disposal area shall be consummated by the County DPW or subsequent holders of any interest in the property without adequate and complete provisions for the full compliance with this Consent Order. Within 30 days of the conveyance of the title of the land within the Phase III disposal area to the County DPW, the County DPW shall prepare and execute a restrictive covenant applicable to the deeds to said disposal area in accordance with Section 16 of the Solid Waste Management Act. The restrictive deed covenant shall state that (a) the property has been used for the disposal of waste substances, and (b) the use of the property must not be allowed to disturb or adversely affect the integrity of any required containment system, treatment system, or monitoring system installed pursuant to this Consent Order. The County DPW shall promptly record the restrictive deed covenant and submit a copy of the recorded instrument to the Department. The restriction shall remain recorded unless the County DPW or a subsequent holder of any interest in the property shall demonstrate to the Department that any required containment, treatment, or monitoring systems are no longer reasonably necessary to prevent a release of a hazardous substance to the environment.

Water Supply Wells

33. In the event the Michigan Department of Public Health (MDPH) reasonably determines that the operation of the disposal site has contaminated any domestic water supply well, the County DPW agrees to provide an alternative source of potable water, capable of meeting all normal domestic needs, as determined by the MDPH, to any resident whose well is thus affected, until such time as the MDPH determines that an alternate source of potable water is no longer necessary.

Right of Entry

34. The parties agree that the Department, its agents, employees, and representatives shall have the right to enter upon the existing and new landfill areas at all reasonable times for the purpose of inspecting, photographing, testing, and monitoring the surface and the sub-surface areas of the disposal site, removing soil and water samples, and determining compliance with the provisions of this Consent Order and any applicable State laws. The Department agrees to provide the County DPW with copies of any and all testing results completed or compiled by its agents, employees, or representatives. This paragraph does not limit the authority of the Department to inspect, photograph, test, or monitor at the disposal site pursuant to any statutory authority.

Reservation of Rights

35. The parties agree that the issue of resource damage has not been completely addressed by the execution of this Consent Order. It is agreed that the State of Michigan does not waive the right to bring an appropriate action to recover for resource damages that are not remediated by the County DPW.
36. The parties further agree that the Department may hereafter institute civil action against the County DPW for the purpose of enforcing this Consent Order.
37. The parties agree that execution of this Consent Order shall not be construed to waive, estop, or otherwise diminish the Department's right to seek or impose civil liability upon, and seek appropriate civil relief from, the County DPW or any other person for degradation of waters of the State and the designated uses thereof arising out of their ownership, operation, and/or use of the disposal site, provided, however, that the Department shall not institute any civil action against the County DPW for degradation of waters if the County DPW complies with this Consent Order and successfully remediates any groundwater contamination resulting from operation of the disposal site pursuant to this Consent Order.

Force Majeure

38. This agreement shall be performed within the time limitations set forth in this agreement unless such performance is delayed by events which would constitute excusable delay. Excusable delay is delay in the completion, scheduling, or performance of a particular task which arises from reasonably unforeseeable causes beyond the reasonable control and

intentional fault of the County DPW. The County DPW shall take all reasonable measures to avoid and to minimize the effect of any event which causes any delay, including an excusable delay, in the timely performance of this Consent Order. If any event occurs which delays, prevents, or leads the County DPW to anticipate delays or prevention of meeting the scheduled dates of any term or condition of this agreement, the County DPW shall give notice to the Department of the event by 5:00 p.m. of the next business day.

Enforcement

39. The parties agree that the terms and conditions of this order shall be enforceable in a State of Michigan Circuit Court.

STATE OF MICHIGAN
DEPARTMENT OF NATURAL RESOURCES

By: 
David F. Hales, Director

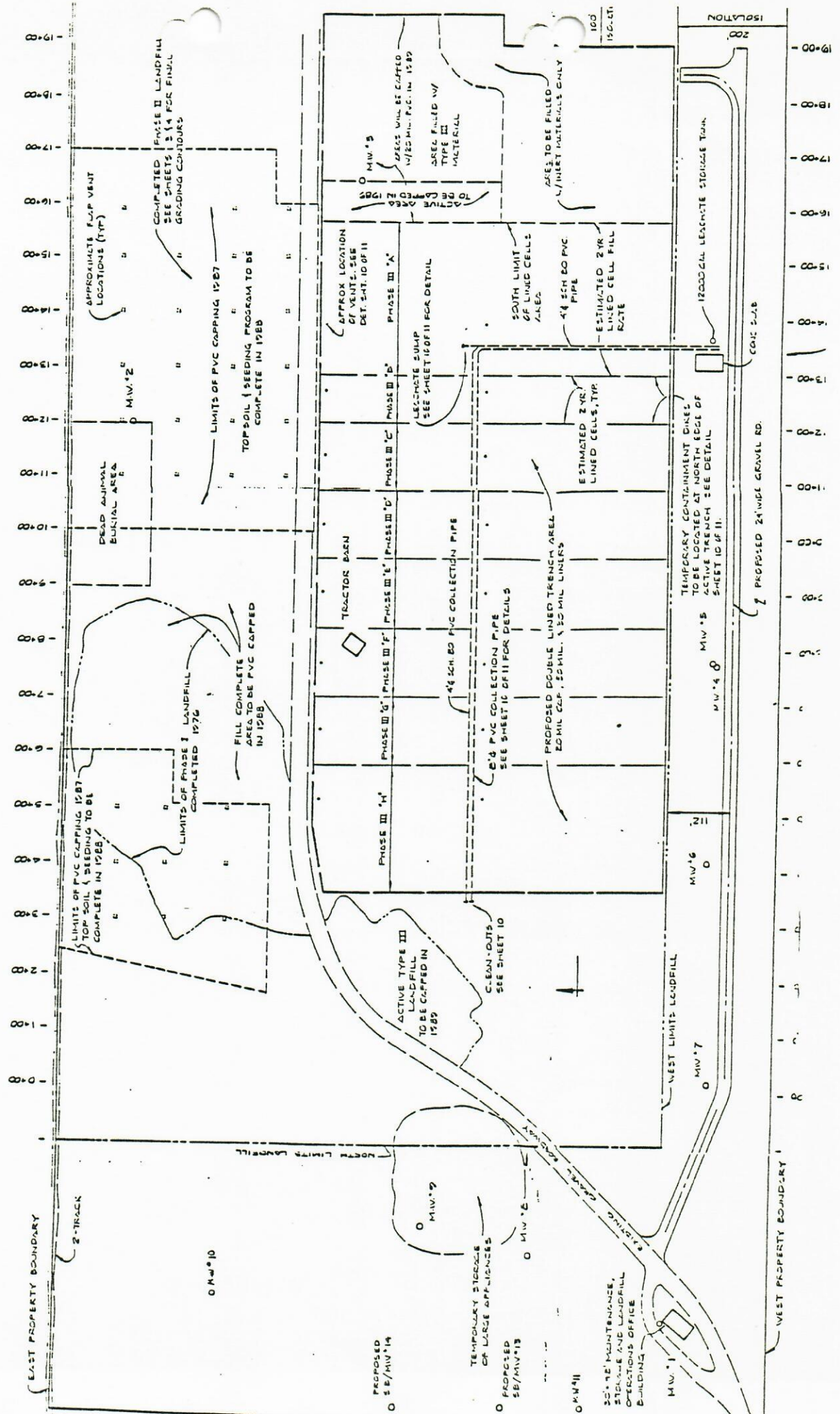
Date: 05/25/89

WEXFORD COUNTY DEPARTMENT OF
PUBLIC WORKS

By: Frank J. Ziska
Frank J. Ziska, Chairman Wexford
County Board of Public Works

Date: May 16, 1989

ATTACHMENT A



ATTACHMENT B

HYDROGEOLOGIC STUDY PLAN

1. Based on existing hydrogeologic data, the horizontal and vertical extent of any contaminated groundwater emanating from the Wexford County DPW disposal site will be determined as follows:
 - a. Groundwater quality at monitor well #1 will be verified by removing the pump, dropping pipe, and resampling the well for those organic constituents listed on Attachment C.
 - b. New soil borings will be drilled downgradient of the disposal site at those locations shown in Figure 1, areas suspected of lying within the contaminant plume. These initial soil borings will be completed using hollow stem augers with a screened auger lead section. Groundwater samples will be obtained at 10 foot intervals to a depth of 250 feet below land surface. If this sampling process indicates a plume is deeper than 250 feet below land surface, the borings will be completed using cable tool methods. The water samples will be analyzed for those parameters listed in Attachment C. One or both of these soil borings will be permanently cased as wells if existing wells are found to not be representative of the contaminant plume.
 - c. Based on soil borings and water quality data collected under paragraphs a and b, the forward edge of the plume will be estimated. Additional soil borings will be drilled no more than 900 feet downgradient of those soil borings listed in paragraph b. This process will be repeated until water quality data shows no detectable concentrations of those organic constituents listed in Attachment C, and conductivity does not exceed 250 umhos. Selected soil borings outside the contaminant plume will be permanently cased as wells.
 - d. In the event the contaminant plume is determined to be within 100 feet of a potable water supply well, a permanent well to define the plume will be located no less than 100 feet from the water supply well.
 - e. Additional borings or wells will be drilled, as reasonably necessary, to determine the extent of any plume of contamination emanating from the disposal site.
2. All borings will be drilled using a drilling method capable of allowing vertical soil and water sampling on ten foot intervals within the zone of saturation for conductivity, temperature, and those organic constituents listed in Attachment C (cable tool or hollow stem auger). Drilling will continue until bedrock is reached or vertical sampling determines that the concentration of organic constituents listed in Attachment C are nondetectable and conductivity does not exceed 250 umhos.
3. All field equipment and well materials shall be steam cleaned prior to use and between each boring.

4. If rotary drilling muds are used, only fresh water will be used as a cutting fluid during drilling.
5. All borings will be logged for lithologic changes using either a baler (if a cable tool is used) or split spoons (if a hollow stem auger is used).
6. The upper portion of each bore hole will be sealed to prevent infiltration from the surface.
7. Permanent wells will be constructed of galvanized steel.
8. Permanent well screens will be set at the depth of highest contamination. Well screens will be three to eight feet long and will be constructed of stainless steel.
9. All permanent wells will be developed using the surge block method.
10. All permanent wells shall have a vented cap to prevent condensation.
11. After installation, all permanent wells will be sampled for those constituents listed in Attachment C. Sampling equipment, sample handling procedures, decontamination procedures, chain of custody procedures, and quality assurance/quality control techniques will be consistent with the Resource Conservation and Recovery Act Groundwater Monitoring Technical Enforcement Guidance Document dated September 1986, and shall be documented.
12. Samples will be analyzed by the attached laboratory tests.
13. Static water levels for each well will be determined using the wet tape method accurate to .01 foot and will be referenced to USGS datum. Stabilized water levels will be obtained from all wells at least 24 hours after well development and within 12 hours of each other.
14. At the completion of data collection, a hydrogeological report shall be prepared which does all of the following:
 - a. Summarizes work done and identifies depth to groundwater and vertical and horizontal groundwater flow directions.
 - b. Summarizes all groundwater quality data obtained to date, referenced to base maps.
 - c. Includes a series of geologic cross-sections passing through representative borings or wells and illustrating existing topography, stratigraphy, stabilized water level readings, and vertical sampling data.
 - d. Includes a water table map based on stabilized water level readings.
 - e. Includes a copy of all well logs, water quality data and other data, as well as documentation of sampling equipment, sample handling procedures, decontamination procedures, chain of custody procedures and quality assurance/quality control procedures used during sampling.

LABORATORY TEST METHODS

Inorganics:

Chlorides	Titration Method Ref: APHA Standard Methods, 14th Ed., 304 (1975)
Sulfates	Turbidimetric Method Ref: APHA Standard Methods, 14th Ed., 496 (1975)
Dissolved Sodium	Atomic Absorption Spectroscopy Method Using Direct Aspiration Ref: APHA Standard Methods, 14th Ed., 148 (1975)
pH	Electrode Method Ref: APHA Standard Methods, 14th Ed., 460 (1975)
Specific Conductance Direct Measurement Method	
Chemical Oxygen Demand	Dichromate Reflux Method Ref: APHA Standard Methods, 14th Ed., 550 (1975)
Dissolved Iron	Atomic Absorption Spectroscopy Method Using Direct Aspiration Ref: APHA Standard Methods, 14th Ed., 148 (1975)
Ammonia Nitrogen	Nessler Method Ref: APHA Standard Methods, 14th Ed., 412 (1975)
Dissolved Chromium	Atomic Absorption Spectroscopy Method Using Extraction/Concentration Ref: APHA Standard Methods, 14th Ed., 151 (1975)
Dissolved Cadmium	Atomic Absorption Spectroscopy Method Using Preconcentration Ref: APHA Standard Methods, 14th Ed., 151 (1975)

Wilcox Associates

- Dissolved Potassium Atomic Absorption Spectroscopy Method
Using Direct Aspiration
REF: APHA Standard Methods, 14th Ed.,
148 (1975)
- Alkalinity Titration Method
REF: APHA Standard Methods, 16th Ed.,
278 (1975)
- Dissolved Lead Atomic Absorption Spectroscopy Method
Using Extraction/Concentration
Ref: APHA Standard Methods, 14th Ed.,
151 (1975)
- Dissolved Magnesium Atomic Absorption Spectroscopy Method
Using Direct Aspiration
REF: APHA Standard Methods, 14th Ed.,
148 (1975)
- Dissolved Calcium Atomic Absorption Spectroscopy Method
Using Direct Aspiration
REF: APHA Standard Methods, 14th Ed.,
148 (1975)
- Total Organic Carbon Combustion-Infrared Method
REF: APHA Standard Methods, 16th Ed.,
508 (1985)
- Dissolved Copper Atomic Absorption Spectroscopy Method
Using Direct Aspiration
REF: APHA Standard Methods, 14th Ed.,
148 (1975)
- Dissolved Nickel Atomic Absorption Spectroscopy Method
Using Direct Aspiration
REF: APHA Standard Methods, 14th Ed.,
148 (1975)
- Dissolved Zinc Atomic Absorption Spectroscopy Method
Using Direct Aspiration
REF: APHA Standard Methods, 14th Ed.,
148 (1975)

Organics:

- Scan 1 EPA Method 601
EPA Test Methods, Federal Register
Vol. 49, No. 209
- Scan 2 EPA Method 602
EPA Test Methods, Federal Register
Vol. 49, No. 209

ATTACHMENT C
MONITORING PARAMETERS

Inorganics, Metals, and Indicators:

Total Organic Carbon (TOC), Chemical Oxygen Demand (COD), Conductivity, Ammonia, Alkalinity, Chloride, Sulfate, Dissolved Calcium, Dissolved Potassium, Dissolved Sodium, Dissolved Magnesium, Dissolved Cadmium, Dissolved Chromium, Dissolved Copper, Dissolved Iron, Dissolved Nickel, Dissolved Lead, Dissolved Zinc, pH, and Static Water Levels.

Purgeable Halocarbons

Bromodichloromethane	1,2 - Dichloroethene (cis & trans)
Bromoform	1,2 - Dichloropropane
Carbon Tetrachloride	1,3 Dichloropropane (cis & trans)
Chlorobenzene	Methylene Chloride
Chloroform	1,1,2,2 - Tetrachloroethane
Dibromochloromethane	Tetrachloroethene
1,1 - Dichloroethane	1,1,1 - Trichloroethane
1,2 - Dichloroethane	1,1,2 - Trichloroethane
1,1 - Dichloroethene	Trichloroethene

Purgeable Aromatic Hydrocarbons

Benzene	Toluene
Ethylbenzene	Xylene Isomers (o, m, and p)